

Thm. Let E be an Extension of a field F , $\alpha, \beta \in E$ be Algebraic over F , with $\deg(\alpha, F) = n$.

α, β are conjugate over F if and only if: a Conjugation map

$$\psi_{\alpha, \beta}: F(\alpha) \rightarrow F(\beta): \sum_{i=0}^{n-1} C_i \cdot \alpha^i \mapsto \sum_{i=0}^{n-1} C_i \cdot \beta^i$$

is Isomorphism.

Proof. First, suppose that α, β are Conjugate over F with respect to $p(x) \in F[x]$.

Now, the evaluation Homomorphisms

$$\phi_\alpha: F[x] \rightarrow F(\alpha): f(x) \mapsto f(\alpha)$$

$$\phi_\beta: F[x] \rightarrow F(\beta): f(x) \mapsto f(\beta)$$

have a same kernel $(p(x))$, because

$$f(x) \in (p(x)) \Leftrightarrow f(\alpha) = 0 \Leftrightarrow p(x) \mid f(x).$$

These induce Natural Field Isomorphisms:

$$\phi_\alpha: F[x]/(p(x)) \rightarrow F(\alpha): f(x) + (p(x)) \mapsto f(\alpha)$$

$$\phi_\beta: F[x]/(p(x)) \rightarrow F(\beta): f(x) + (p(x)) \mapsto f(\beta)$$

Define an Isomorphism $\psi_{\alpha, \beta}$ as Composition of the Natural Isomorphisms:

$$\psi_{\alpha, \beta}: F(\alpha) \rightarrow F(\beta): x \mapsto (\phi_\beta \circ \phi_\alpha^{-1})(x)$$

This $\psi_{\alpha, \beta}$ maps $\sum_{i=0}^{n-1} C_i \alpha^i$ to $\sum_{i=0}^{n-1} C_i \beta^i$, because: $\psi_{\alpha, \beta}(\alpha) = \phi_\beta(\phi_\alpha^{-1}(\alpha)) = \phi_\beta(x) = \beta$ implies

$$\psi_{\alpha, \beta} \left(\sum_{i=0}^{n-1} C_i \alpha^i \right) = \sum_{i=0}^{n-1} C_i \cdot (\psi_{\alpha, \beta}(\alpha))^i = \sum_{i=0}^{n-1} C_i \beta^i.$$

Conversely, suppose that given $\psi_{\alpha, \beta}$ is Isomorphism.

Put $m_{\alpha, F}$, $m_{\beta, F}$ be monic minimal irreducible polynomial of α, β , respectively. Then,

$$\psi_{\alpha, \beta}(m_{\alpha, F}(\alpha)) = m_{\alpha, F}(\beta) = 0$$

$$\psi_{\alpha, \beta}^{-1}(m_{\beta, F}(\beta)) = m_{\beta, F}(\alpha) = 0.$$

These implies $m_{\alpha, F}$ divides $m_{\beta, F}$ and $m_{\beta, F}$ divides $m_{\alpha, F}$. Since both are monic, same.

Corollary. Let $\alpha \in F$ be an algebraic over a field F .

For every Monomorphism $\psi: F(\alpha) \rightarrow \overline{F}$ which fixed F , α and $\psi(\alpha)$ are conjugate.

Conversely, for each conjugate β of α over F , there exists one and only one Monomorphism

$$\psi_{\alpha, \beta}: F(\alpha) \rightarrow \overline{F}: \sum_{i=0}^{\deg \alpha} C_i \alpha^i \mapsto \sum_{i=0}^{\deg \alpha} C_i \beta^i$$

which satisfied $\psi_{\alpha, \beta}(\alpha) = \beta$ and fix F .

Proof. Suppose that $\psi: F(\alpha) \rightarrow \overline{F}$ is a monomorphism which fixed F .

Let $m_{\alpha, F} = \sum_{i=0}^n \alpha_i x^i \in F[x]$ be a monic minimal irreducible polynomial of α . Then,

$$m_{\alpha, F}(\psi(\alpha)) = \sum_{i=0}^n \alpha_i \psi(\alpha^i) = \psi\left(\sum_{i=0}^n \alpha_i \alpha^i\right) = \psi(0) = 0.$$

That is, $\psi(\alpha)$ is a conjugate of α .

Lemma Let R and S be non-zero rings with identity 1_R and 1_S , respectively.
Let $\varphi: R \rightarrow S$ be a non-trivial Ring Homomorphism.

1) If $\varphi(1_R) \neq 1_S$, then $\varphi(1_R)$ is zero divisor in S .

2) If $\varphi(1_R) = 1_S$, then for each unit $u \in R$, $\varphi(u)^{-1} = \varphi(u)$.

Proof. Since φ preserves the multiplication,

$$\begin{aligned}\varphi(1_R) \cdot \varphi(1_R) &= 1_S \cdot 1_S = 1_S = \varphi(1_R) \\ \Rightarrow \varphi(1_R) \cdot (\varphi(1_R) - 1_S) &= 0.\end{aligned}$$

Since $\varphi(1_R)$ is not 1_S , thus $\varphi(1_R)$ is a zero divisor.

To show the second assertion, let a unit $u \in R$ be given. Then

$$\varphi(u) \cdot \varphi(u^{-1}) = \varphi(u \cdot u^{-1}) = \varphi(1_R) = 1_S.$$

Corollary. If $\varphi: R \rightarrow S$ is a Ring-Homomorphism and S is integral domain, then φ preserves an identity.

Lemma. Let K be a field with $\text{ch}(K) = 0$.

Then, the prime subfield of K is isomorphic to $\mathbb{Q}$.

Lemma. Let K be a field and $F \subseteq K$ be a prime subfield.

For any $\sigma \in \text{Aut}(K)$, σ fixes F .

Proof. Trivial.

Lemma. Non-trivial field Homomorphism is injective.

Proof. Let $\varphi: F \rightarrow F'$ be a field Homomorphism.

Since $\text{Ker } \varphi \leq F$ is ideal of field F , $\text{Ker } \varphi$ is either $\{0\}$ or F . $\square$

Note: the codomain is not necessary field, but ring.

Thm. Let E be an algebraic Extension of a field F .

For an Isomorphism $\mathcal{Q}: F \rightarrow F'$, for some $E' \leq \overline{F'}$, there exists an Extension Isomorphism $\phi: E \rightarrow E'$ s.t. $\forall x \in F, \phi(x) = \mathcal{Q}(x)$.

$$\alpha \sim \beta$$

$$F(\alpha) \rightarrow F(\beta)$$

Thm. Let E be a finite extension of F .

Let $\mathcal{Q}: F \rightarrow F'$ be a field Isomorphism. Then,

there exist finite number of extension of $\mathcal{Q}$ to $\phi: E \rightarrow E' \leq \overline{F'}$ and the number of extensions is completely determined by E and F .

Proof. Let two Isomorphisms

$$\mathcal{Q}_1: F \rightarrow F'_1, \quad \mathcal{Q}_2: F \rightarrow F'_2$$

Then, the composition

$$\mathcal{Q}_2 \circ \mathcal{Q}_1^{-1}: F'_1 \rightarrow F'_2$$

$$\downarrow \nearrow$$

$$\begin{array}{ccc} \overline{F'_1} & \xrightarrow{\quad \quad} & \overline{F'_2} \\ | & & | \\ E'_1 \xleftarrow{\phi_1} E & \xrightarrow{\phi_2} & E'_2 \\ | & & | \\ F'_1 \xleftarrow{\mathcal{Q}_1} F & \xrightarrow{\mathcal{Q}_2} & F'_2 \end{array}$$

is Isomorphism from F'_1 onto F'_2 . Then, by the Existence of Extension, there is

$$\psi: \overline{F'_1} \rightarrow \overline{F'_2} \text{ s.t. } \forall x \in F'_1, \psi(x) = (\mathcal{Q}_2 \circ \mathcal{Q}_1^{-1})(x).$$

And, there exist Extensions of $\mathcal{Q}_1$ on F to E ,

$$\phi_1: E \rightarrow E'_1 \leq \overline{F'_1} \text{ s.t. } \forall x \in F, \phi_1(x) = \mathcal{Q}_1(x).$$

Then,

$$\phi_2: E \rightarrow E'_2: x \mapsto \psi(\phi_1(x))$$

for each $x \in E$,

$$\phi_2(x) = \psi(\phi_1(x))$$

$$\phi_2: E \rightarrow E'_2 \leq \overline{F'_2} \text{ s.t. } \forall x \in F, \phi_2(x) = \mathcal{Q}_2(x).$$

Existence of Extension of Isomorphism.

Let $\phi: F \rightarrow F'$ be a field Isomorphism.

If E is an algebraic Extension of F , then there exists an Isomorphism

$$\phi': E \rightarrow E' \subseteq F' \text{ s.t. } \phi|_F = \phi$$

Proof. Define a set:

$$P = \left\{ (L, \lambda) \mid \begin{array}{l} L \text{ is field s.t. } F \subseteq L \subseteq E, \\ \lambda: L \rightarrow F' \text{ is field Homomorphism s.t. } \lambda|_F = \phi \end{array} \right\}$$

Since $(F, \phi) \in P$, P is non-empty. Define order on P as:

$$(L_1, \lambda_1) \leq (L_2, \lambda_2) \Leftrightarrow L_1 \subseteq L_2 \text{ and } \lambda_2|_{L_1} = \lambda_1$$

Let $T = \{ (L_i, \lambda_i) \mid i \in I \}$ be a Chain in P . Put $H \stackrel{\text{def}}{=} \bigcup_{i \in I} L_i$.

To show that H is a subfield of E , let $x, y \in H$ be given.

By definition, $x \in L_{i_1}$ and $y \in L_{i_2}$ for some $i_1, i_2 \in I$,

and since T is a Chain, either $L_{i_1} \subseteq L_{i_2}$ or $L_{i_2} \subseteq L_{i_1}$.

WLOG, assume $L_{i_1} \subseteq L_{i_2}$. Then, $x, y \in L_{i_2}$. Thus, if

$$x \pm y, x \cdot y \in L_{i_2} \subseteq H \text{ and if } y \neq 0, \frac{x}{y} \in L_{i_2} \subseteq H$$

Therefore, H is subfield of E containing F .

Meanwhile, define a map:

$$\lambda: H \rightarrow F'; x \mapsto \lambda_i(x) \text{ if } x \in L_i \text{ for some } i \in I.$$

It is well-defined because: for any $x \in H$, $x \in L_i$ for some $i \in I$ by definition.

And, if $x \in L_i$ and $L_i \subseteq L_j$ where $L_i \leq L_j$ then

$$\lambda_j(x) = \lambda_j|_{L_i}(x) = \lambda_i(x) \Rightarrow ?$$

thus $\lambda(x)$ is uniquely determined. And, it is homomorphism, because:

Let $x, y \in H$ be given. Then, for some $i \in I$, $x, y \in L_i$. Now,

$$\lambda(x+y) = \lambda_i(x+y) = \lambda_i(x) + \lambda_i(y) = \lambda(x) + \lambda(y)$$

$$\lambda(xy) = \lambda_i(xy) = \lambda_i(x) \lambda_i(y) = \lambda(x) \lambda(y)$$

And λ must fix F , thus non-trivial homomorphism.

Finally, (H, λ) is upper bound of T in P .

By Zorn's Lemma, there exists a Maximal element $(K, \Phi) \in \mathcal{P}$.

To show that $K=E$, suppose that $K \neq E$. Choose $\alpha \in E \setminus K$.

Since $\alpha \in E$ is algebraic over F , also algebraic over K .

Let $p(x) \in K[x]$ be a monic irreducible polynomial such that $p(\alpha) = 0$.

Write $K' = \Phi[K] \leq F$ and

$$p(x) = a_0 + a_1x + \dots + a_nx^n.$$

Consider the Canonical Isomorphism

$$\psi_\alpha: K[x]/(p(x)) \rightarrow K(\alpha); f(x) + (p(x)) \mapsto f(\alpha)$$

And

$$\bar{p}(x) = \Phi(a_0) + \Phi(a_1)x + \dots + \Phi(a_n)x^n \in K'[x]$$

Since Φ is an Isomorphism, $\bar{p}$ is irreducible in K' .

Moreover, for root $\alpha' \in F'$ of $\bar{p}$, let

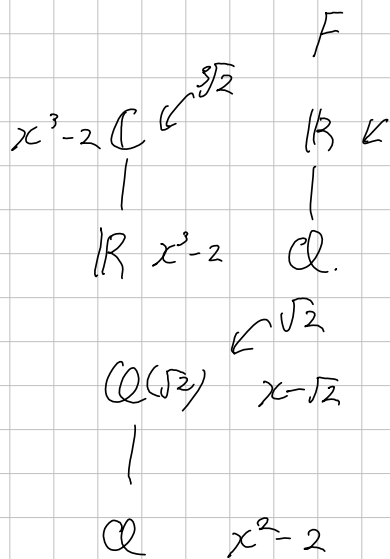
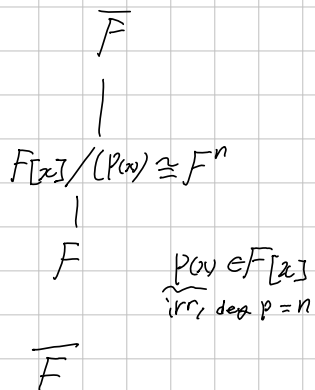
$$\psi_{\alpha'}: K'[x]/(\bar{p}(x)) \rightarrow K'(\alpha'); f(x) + (\bar{p}(x)) \mapsto f(\alpha')$$

is Canonical Isomorphism.

Let

$$\bar{\Phi}: K[x]/(p(x)) \rightarrow K[x]/(\bar{p}(x)); f(x) + (p(x)) \mapsto$$

Summary of field theory.



$$\alpha \in E/K \Rightarrow \exists \underbrace{p(x) \in K[x]}_{\text{irr}} \text{ s.t. } p(\alpha) = 0.$$

$$\begin{array}{ccccc}
 K[x] & \xrightarrow{\sim} & K'[x] & & \\
 \swarrow & & \downarrow & \searrow & \\
 & K & \xrightarrow{\sim} & K' & \\
 \swarrow & & & & \searrow \\
 K(\alpha) & \longleftarrow & K[x]/(p(x)) & \xrightarrow{\quad} & K'[x]/(p'(x)) \longrightarrow K'(\alpha')
 \end{array}$$

$$\begin{array}{ccccc}
 \overline{F_1} & \xrightarrow{\quad} & \overline{F_2} & & \\
 \downarrow & & \downarrow & & \\
 E_1 & \xleftarrow{\overline{\phi_1}} & E & \xrightarrow{\overline{\phi_2}} & E_2 \\
 \downarrow & & \downarrow & & \downarrow \\
 F_1 & \xleftarrow{\phi_1} & F & \xrightarrow{\phi_2} & F_2
 \end{array}$$

$\phi: F \rightarrow F'$ Isomorphism, $E = F(\alpha_1, \dots, \alpha_n)$ where $\alpha_1, \dots, \alpha_n \in \overline{F}$.

If $\text{irr}(\alpha_i, F) = a_{i,0} + a_{i,1}x + \dots + a_{i,m_i}x^{m_i}$,
 then $\overline{\phi}(\alpha_i)$ must be a root of $\phi_{*c}(\text{irr}(\alpha_i, F)) \in F[x]$.

$$\begin{array}{ccc}
 E & \xrightarrow{\overline{\phi}} & E' \\
 \downarrow & & \downarrow \\
 F & \xrightarrow{\phi} & F' \\
 \swarrow & & \searrow \\
 F[x] & \xrightarrow{\phi_{*c}} & F'[x]
 \end{array}$$

Thm. Let E be a finite extension of F . Let $\mathcal{Q}: F \rightarrow F'$ be a field Isomorphism.

Then, the number of extensions of $\mathcal{Q}$ to E is finite.

Rigorously, $\{\overline{\mathcal{Q}}: E \rightarrow \overline{F'} \mid \overline{\mathcal{Q}}|_F = \mathcal{Q}, \overline{\mathcal{Q}} \text{ is field homomorphism}\}$ is finite.

Proof. Since E is finite dimension over F , there exist $\alpha_1, \dots, \alpha_n \in E$ s.t.

$$E = F(\alpha_1, \dots, \alpha_n).$$

For each $1 \leq i \leq n$, let

$$\text{irr}(\alpha_i, F) = a_{i,0} + a_{i,1}x + \dots + a_{i,m_i}x^{m_i} \in F[x]$$

If $\overline{\mathcal{Q}}: E \rightarrow \overline{F'}$ is a homomorphism s.t. $\overline{\mathcal{Q}}|_F = \mathcal{Q}$, then $\overline{\mathcal{Q}}(\alpha_i)$ must be a root of:

$$\text{Because } \mathcal{Q}(a_{i,0}) + \mathcal{Q}(a_{i,1})x + \dots + \mathcal{Q}(a_{i,m_i})x^{m_i} \in F[x]$$

$$\sum_{k=0}^{m_i} \mathcal{Q}(a_{i,k}) (\overline{\mathcal{Q}}(\alpha_i))^k = \overline{\mathcal{Q}} \left(\sum_{k=0}^{m_i} a_{i,k} \alpha_i^k \right) = \overline{\mathcal{Q}}(0) = 0.$$

Therefore, there exists possibility of numbers of

$$\deg \text{irr}(\alpha_1, F) \cdot \deg \text{irr}(\alpha_2, F) \cdots \deg \text{irr}(\alpha_n, F)$$

More specifically, note that: E/F is finite extension $\Leftrightarrow E = F(\alpha_1, \dots, \alpha_n)$ where $\alpha_i \in E$.

Let $\alpha \in \overline{F}$ be an algebraic over F . Then there exists a monic irreducible

$$f(x) = a_0 + a_1x + \dots + a_{n-1}x^{n-1} + x^n \in F[x] \quad (n \geq 1)$$

satisfying $f(\alpha) = 0$. Let $\mathcal{Q}: F \rightarrow F'$ be an Isomorphism. We have shown that:

$$\overline{\mathcal{Q}}: F(\alpha) \rightarrow \overline{F'} \text{ s.t. } \overline{\mathcal{Q}}|_F = \mathcal{Q}, \overline{\mathcal{Q}} \text{ is field homomorphism.}$$

exists. Since the fact that $F(\alpha) = \langle \{1, \alpha, \dots, \alpha^{n-1}\} \rangle_F$,

$$F(\alpha) = \overline{\mathcal{Q}}[F(\alpha)]$$

For any $\sum_{i=0}^{n-1} a_i \alpha^i \in F(\alpha)$,

$$\begin{array}{c} 1 \quad 1 \\ F \rightarrow F' \end{array}$$

$$\overline{\mathcal{Q}} \left(\sum_{i=0}^{n-1} a_i \alpha^i \right) = \sum_{i=0}^{n-1} \mathcal{Q}(a_i) \cdot \overline{\mathcal{Q}}(\alpha)^i \in \overline{\mathcal{Q}}[F(\alpha)] \subseteq \overline{F'}.$$

That is, $\overline{\mathcal{Q}}$ is determined by value of $\overline{\mathcal{Q}}(\alpha)$ in $\overline{F'}$.

Actually, $\overline{\mathcal{Q}}(\alpha)$ is the one of root of a polynomial

$$\overline{\mathcal{Q}}_x(\text{irr}(\alpha, F)) \in \overline{F'}[x] \text{ where } \overline{\mathcal{Q}}_x: F[x] \rightarrow \overline{F'}[x]: \sum a_i x^i \mapsto \sum \mathcal{Q}(a_i) x^i$$

Consider an Automorphism:

$$\gamma: \mathbb{Q}(\sqrt{3}) \rightarrow \mathbb{Q}(\sqrt{3}); a_0 + a_1\sqrt{3} \mapsto a_0 - a_1\sqrt{3}$$

And extension of γ to $\mathbb{Q}(\sqrt{3}, i)$

$$\overline{\gamma}: \mathbb{Q}(\sqrt{3}, i) \rightarrow \overline{\mathbb{Q}}$$

Since $\text{irr}(i, \mathbb{Q}(\sqrt{3})) = x^2 + 1$, the extension $\overline{\gamma}$ is determined by a value of $\overline{\gamma}(i) = i$ or $-i$.

$$\overline{\gamma}: \mathbb{Q}(\sqrt{3}, i) \rightarrow \overline{\mathbb{Q}}; a_0 + a_1 i \mapsto \gamma(a_0) \pm \gamma(a_1) \cdot i$$

Consider an Automorphism:

$$\mathcal{Q}: \mathbb{Q}(\pi) \rightarrow \mathbb{Q}(\pi) \text{ s.t. } \mathcal{Q}(\pi) = -\pi$$

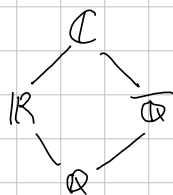
$$\mathcal{Q}\left(\frac{f(x)}{g(x)}\right) = \frac{f(\mathcal{Q}(x))}{g(\mathcal{Q}(x))}$$

$$\mathbb{Q}(\pi)$$

$$\begin{aligned} \mathcal{Q}(\pi^2) &= \mathcal{Q}(\pi)^2 \\ &= (-\pi)^2 \\ &= \pi^2 \end{aligned}$$

$$\begin{array}{c} \mathbb{Q}(\pi) \\ | \\ \mathbb{Q}(\pi^2) \\ | \\ \mathbb{Q} \end{array}$$

$$\left\{ \frac{f(x)}{g(x)} \mid f, g \in F[x], g(x) \neq 0 \right\} = F(x).$$



Let α, β be transcendental over F . Define a map:

$$\mathcal{Q}: F(\alpha) \rightarrow F(\beta); \frac{f(\alpha)}{g(\alpha)} \mapsto \frac{f(\beta)}{g(\beta)}$$

Then $\mathcal{Q}$ is an Isomorphism.

$$\mathbb{Q}(\pi)(\sqrt{\pi}) \quad x^2 - \pi$$

$$\overline{\mathcal{Q}}: \mathbb{Q}(\sqrt{\pi}) \rightarrow \overline{\mathbb{Q}(\pi)}; a_0 + a_1\sqrt{\pi} \mapsto \mathcal{Q}(a_0) + \mathcal{Q}(a_1)\overline{\mathcal{Q}(\sqrt{\pi})}$$

$$a_0, a_1 \in \mathbb{Q}(\pi)$$

where $\overline{\mathcal{Q}(\sqrt{\pi})}$ is a root of $x^2 + \pi$

Thus, $\overline{\mathcal{Q}(\sqrt{\pi})} = i\sqrt{\pi}$ or $-i\sqrt{\pi}$

Let E be an Extension of F . Let $t \in E$ be an indeterminate.

Consider a simple extension $F(t)$.

If $t \in E$ is transcendental over F , that is,

For any non-zero polynomial $p(x) \in F[x]$ $p(t) \neq 0$

$$\text{Then, } F(t) = \left\{ \frac{f(t)}{g(t)} \mid f, g \in F[x], g \neq 0 \right\}$$

If $t \in E$ is algebraic over F ,

then there exists an monic irreducible polynomial $p(x) \in F[x]$ s.t. $p(t) = 0$. Now,

$$F(t) = \left\{ \frac{f(t)}{g(t)} \mid f(x) + (p(x)), g(x) + (p(x)) \in F[x]/(p(x)), g(x) + (p(x)) \neq (p(x)) \right\}$$

$$g(t) \neq 0 \Leftrightarrow g(x) \notin (p(x))$$

Thm. Let E be an Extension field of F . Let $t \in E$ be a given as indeterminate.

The smallest subfield of E containing F and t is

$$F(t) = \left\{ \frac{f(t)}{g(t)} \mid f, g \in F[x], g(t) \neq 0 \right\}$$

$$= \begin{cases} \left\{ \frac{f(t)}{g(t)} \mid f, g \in F[x], g \neq 0 \right\} & \text{if } t \text{ is transcendental over } F \\ \left\{ \frac{f(t)}{g(t)} \mid f, g \in F[x], g(t) \neq 0 \right\} & \text{if } t \text{ is algebraic over } F \end{cases}$$

|||
 $F[x]/(p(x))$ where $p(x)$ is irreducible in $F[x]$ having t as a root.

$$g(t) \neq 0 \Leftrightarrow g(x) \notin (p(x))$$

$$F[x] \rightarrow \cdot ; f(x) \mapsto f(t)$$

$$\ker \phi = (p(x))$$

Thm. Let E be an algebraic extension of F .

Let $\mathcal{Q}_1: F \rightarrow F_1$ and $\mathcal{Q}_2: F \rightarrow F_2$ be field isomorphisms.

Then, there exists a bijection between

$$S_1 = \left\{ \widehat{\mathcal{Q}}_1: E \rightarrow F_1 \mid \widehat{\mathcal{Q}}_1 \text{ is field homomorphism, } \widehat{\mathcal{Q}}_1|_F = \mathcal{Q}_1 \right\}$$



$$S_2 = \left\{ \widehat{\mathcal{Q}}_2: E \rightarrow F_2 \mid \widehat{\mathcal{Q}}_2 \text{ is field homomorphism, } \widehat{\mathcal{Q}}_2|_F = \mathcal{Q}_2 \right\}$$

proof. Let $\Phi: F_1 \rightarrow F_2$ be a Extension Isomorphism of $\mathcal{Q}_2 \circ \mathcal{Q}_1^{-1}$.

Let $\widehat{\mathcal{Q}}_1 \in S_1$ be given. Then, a map

$$\begin{array}{ccc} \Phi \widehat{\mathcal{Q}}_1: E & \rightarrow & F_2 : a \mapsto \Phi(\widehat{\mathcal{Q}}_1(a)) \\ \widehat{\mathcal{Q}}_1 \downarrow & & \uparrow \Phi \\ & F_1 & \end{array}$$

is field Homomorphism s.t. $\Phi \widehat{\mathcal{Q}}_1|_F = \mathcal{Q}_2$ because: for any $a \in F$,

$$\Phi(\widehat{\mathcal{Q}}_1(a)) = \Phi(\mathcal{Q}_1(a)) = (\mathcal{Q}_2 \circ \mathcal{Q}_1^{-1})(\mathcal{Q}_1(a)) = \mathcal{Q}_2(a)$$

Apply same process reversely with respect to Φ^{-1} , then we obtain $\dashv$ correspondence.

Let K be a field, and $K(x)$ be a field of fractions of $K[x]$, i.e.,

$$K(x) = \left\{ \frac{p(x)}{q(x)} \mid p, q \in K[x], q(x) \neq 0 \right\}$$

Then, $\text{Aut}(K(x)/K)$

$$\mathcal{Q}: K(x) \mapsto K(x);$$

Thm. Let $F(x)$ be a field of fractions of $F[x]$.

φ is an Automorphism of $F(x)$ which fixes F .

$$\varphi: F(x) \rightarrow F(x): f(x)/g(x) \mapsto f\left(\frac{ax+b}{cx+d}\right)g\left(\frac{ax+b}{cx+d}\right)^{-1}$$

for some $a, b, c, d \in F$, $ad - bc \neq 0$

Proof. The upper direction is clear, Just Calculation.

Since a Homomorphism on $F(x)$ onto $F(x)$ which fixes F is determined by $\varphi(x)$, find the possible value of $\varphi(x)$. Denote

$$\varphi(x) = \frac{f(x)}{g(x)}$$

where $f, g \in F[x]$, f and g are relatively prime.

If $\max\{\deg f, \deg g\} = 0$, that is, both are constant, then $\varphi[F(x)] = F \neq F(x)$. Contradiction. Suppose $\max\{\deg f, \deg g\} \geq 1$. If larger than 1, then

$$[F(x):F(\varphi(x))] > 1$$

because $f(x) - \frac{f(x)}{g(x)}g(x) \in F\left(\frac{f(x)}{g(x)}\right)[x]$ is irreducible, by the Gauss Lemma,

(since x is indeterminate, $f(x)/g(x)$ also has no root in F).

Therefore, for $t = \frac{f(x)}{g(x)}$, $f(x) - tg(x) \in F[t][x] = F[x][t]$ is irreducible,

so that it is irreducible in $F(t)[x]$ and has x as a root.

It is contradiction. Now, $\varphi(x) = \frac{ax+b}{cx+d}$ is the unique possibility, with $ad - bc \neq 0$.

Thm. Let F be a field, and E be an algebraic extension. That is, $F \subseteq E \subseteq \bar{F}$.

E is a splitting field over F
iff

If $\sigma: F \rightarrow \bar{F}$ is automorphism which fixes F , then $\sigma[E] = E$

Proof. Let E be a splitting field over F in $\bar{F}$ of $\{f_i(x) \mid i \in I\}$,
and $\sigma: F \rightarrow \bar{F}$ be an Automorphism which fixes F .

Let $S = \{a \in \bar{F} \mid f_i(a) = 0 \text{ for some } i \in I\}$

Define $\bar{S} = \bigcap_{\substack{F \subseteq K \subseteq \bar{F} \\ S \subseteq K}} K$ (the smallest subfield of $\bar{F}$ containing F and S)

Then, by definition, $\bar{S} = E$.

Since σ fixes F and elements of $\bar{S} = E$ are represented as:

finite sum of finite product of $\sum_{k=0}^{\deg f_i - 1} a_k \sigma^k$, $\sigma \in S$, $a_k \in F$

σ is determined by the value $\sigma(a)$ for each $a \in S$

But, $\sigma(a)$ is a root of $\text{irr}(a, F)$ (because σ fixes F), thus $\sigma(a) \in E$.

This implies $\sigma[E] \subseteq E$.

$$\begin{array}{ccc}
 \bar{F} & \longrightarrow & \bar{F} \\
 | & & | \\
 F(\alpha) & \longrightarrow & F(\beta) \\
 | & & | \\
 F & \longrightarrow & F
 \end{array}$$

Proof

Thm. Let F be a field, and let x be an indeterminate over F .

Then, the field of fractions of $F[x]$ is the smallest extension of F containing x .
Moreover, for any transcendental element t over F ,

$$F(x) \cong F(t)$$

Proof. The first assertion is clear. And, a map

$$\varphi: F(x) \rightarrow F(t): \frac{f(x)}{g(x)} \mapsto \frac{f(t)}{g(t)}$$

gives the Isomorphism.

Thm. Let F be a field, x be an indeterminate over F .

Let $\frac{p(x)}{g(x)} \in F(x)$ s.t. $g(x) \neq 0$, $\gcd(p(x), g(x)) = 1$. Denote $E = F\left(\frac{p(x)}{g(x)}\right) \subseteq F(x)$

1). $p(x) - \frac{p(x)}{g(x)} \cdot g(x) \in E[x]$ is irreducible over E and has x as a root.

Proof. Denote $\frac{p(x)}{g(x)} = t$. By the Gauss Lemma,

$p(x) - tg(x)$ is irreducible over $F(t)$ iff $p(x) - tg(x)$ is irreducible over $F[t]$

And, $F[t][x] = F[x][t]$, $p(x) - tg(x)$ is irreducible over $F[x][t]$

having x as a root is clear.

$$\text{if } \frac{p(x)}{g(x)} \text{ a root of } \sum_{i=0}^{n-1} a_i x^i,$$

$$\sum_{i=0}^{n-1} a_i \left(\frac{p(x)}{g(x)} \right)^i = 0$$

$$\Rightarrow \sum_{i=0}^{n-1} a_i \cdot g(x)^{n-1-i} p(x)^i = 0$$

Thm. Let α be algebraic over F , with $\deg(\alpha, F) = n$.

Then, there are at most n different isomorphisms on $F(\alpha)$ into $\bar{F}$ leaving F fixed.

Proof. Let $\varphi: F(\alpha) \rightarrow \bar{F}$ be an isomorphism leaving F fixed.

Then, for arbitrary element $\sum_{i=0}^{n-1} a_i \alpha^i \in F(\alpha)$,

$$\varphi\left(\sum_{i=0}^{n-1} a_i \alpha^i\right) = \sum_{i=0}^{n-1} a_i \varphi(\alpha)^i$$

That is, the φ is determined by the value of $\varphi(\alpha) \in \bar{F}$.

Meanwhile, since φ fixes F ,

$$\varphi(\text{irr}(\alpha, F)(\alpha)) = \text{irr}(\alpha, F)(\varphi(\alpha)) = 0.$$

That is, $\varphi(\alpha)$ must be a root of $\text{irr}(\alpha, F)$; thus there exists only n possibilities.

Thm. Let $F(\alpha_1, \dots, \alpha_n)$ be a finitely generated field of F by $\alpha_1, \dots, \alpha_n$.

Every Automorphism φ of $F(\alpha_1, \dots, \alpha_n)$ leaving F fixed is determined by the values $\varphi(\alpha_1), \dots, \varphi(\alpha_n)$.

Proof. Using mathematical induction for n .

If $n=1$, then for any $f, g, g(\alpha) \neq 0$,

$$\varphi\left(\frac{f(\alpha)}{g(\alpha)}\right) = \frac{f(\varphi(\alpha))}{g(\varphi(\alpha))}$$

Thm. Let E/F be an algebraic extension of a field F .

Let $S \subseteq \text{Aut}(E/F)$. Then $E_S = \overline{E}_{\langle S \rangle}$

Proof. Clearly, $E_S \supseteq \overline{E}_{\langle S \rangle}$. Let $a \in E_S$.

That is, for all $\sigma \in S$, $\sigma(a) = a$

Meanwhile, let $\varphi \in \langle S \rangle$ be given. Then,

$$\varphi = \sigma_1 \circ \dots \circ \sigma_n \text{ for some finite } \sigma_i \in S$$

Then $\varphi(a) = \sigma_1(\dots(\sigma_n(a)) = \dots = a$.

In sum, $\forall a \in E_S$, $\forall \varphi \in \langle S \rangle$, $\varphi(a) = a$. $\therefore a \in \overline{E}_{\langle S \rangle}$.

Thm. $\varphi: \mathbb{Q}(\pi) \rightarrow \mathbb{Q}(\pi)$ s.t. $\varphi(\pi) = -\pi$.

$\mathbb{Q}(\pi)_{\text{inv}} = \mathbb{Q}(\pi^2)$, because $\varphi(\pi^2) = \varphi(\pi)^2 = (-\pi)^2 = \pi^2$

$\overline{\varphi}: \mathbb{Q}(\sqrt{\pi}) \rightarrow \mathbb{Q}(\pi)$ s.t. $\overline{\varphi}|_{\mathbb{Q}(\pi)} = \varphi$

Since $\text{irr}(\sqrt{\pi}, \mathbb{Q}(\pi)) = x^2 - \pi$,

$\overline{\varphi}(\sqrt{\pi})$ is root of $x^2 - \pi$.

$$\overline{\varphi}(\sqrt{\pi})^2 - \varphi(\pi) = 0.$$

Thm. Let F be a field, and let x be an indeterminate over F . Then,

φ is an Automorphism on $F[x]$ leaves F fixed



$$\varphi: F(x) \rightarrow F(x):$$

$$\varphi(x) = \frac{f(x)}{g(x)} \text{ where } \gcd(g(x), f(x)) = 1, g(x) \neq 0$$

If $\max\{\deg f, \deg g\} = n > 1$, then

$$[F(x) : F(\frac{f(x)}{g(x)})] > 1 \text{ because } \forall a \in F(\frac{f(x)}{g(x)}), x - a \neq 0$$

$$x - \frac{ax+b}{cx+d} = 0? \quad ad - bc \neq 0$$

$$\frac{ax+b}{cx+d}$$

$$\frac{\frac{a}{c}x + \frac{b}{c}}{x + \frac{d}{c}}$$

$$x + \frac{d}{c}$$

$$\frac{a}{c} \cdot \left(\frac{x + \frac{b}{a}}{x + \frac{d}{c}} \right)$$

$$ad \neq bc$$

$$\frac{d}{c} \neq \frac{b}{a}$$

$$\frac{x+2}{x+1}$$

$$x+1-1$$

$$\frac{x+2}{x+1} = 1 + \frac{1}{x+1}$$

Thm. If K is algebraically closed,

then any Isomorphism $\varphi: K \rightarrow K(\overline{K})$ s.t. K is algebraic over K' is onto map.

Proof. Consider an extension of $\varphi^{-1}: K' \rightarrow K$ to $\overline{K}$,

$$\overline{\varphi^{-1}}: \overline{K} \rightarrow \overline{K} \in K(\overline{K}) \text{ s.t. } \overline{\varphi^{-1}}|_{K'} = \varphi^{-1}.$$

Then, $\overline{\varphi^{-1}}[K'] = \varphi^{-1}[K'] = K$, thus it is Isomorphism onto K .

$$\text{Now, } \varphi[K] = \varphi[\overline{\varphi^{-1}}[K]] = \varphi[\overline{\varphi^{-1}}[K']] = \varphi[\varphi^{-1}[K']]$$

$$K' = \varphi[\varphi^{-1}[K']]$$

$$K' = \varphi[K] = (\overline{\varphi^{-1}})^{-1}[K]$$

Let $y \in K'$. Then, for some $x \in K$, $\varphi(x) = y$.

Let $\alpha \in K$ be given, since K is algebraic over K' ,

for some $\alpha_i \in K'$, $\sum \alpha_i \alpha^i = 0$.

And, since $\alpha_i \in K' = \varphi[K]$, $\alpha_i = \varphi(b_i)$ for some $b_i \in K$.

Now, $\sum \varphi(b_i) \varphi(\alpha^i) = \varphi(\sum \alpha_i \alpha^i) = 0$. That is, $[K:K'] = 1$.

then every endomorphism $\varphi: K \rightarrow K$ s.t. K is algebraic over $\varphi[K]$ is onto map.

Proof. Since every field homomorphism having non-trivial Image is injective

$\varphi: K \rightarrow \varphi[K]; x \mapsto \varphi(x)$ is Isomorphism.

Now, consider the extension of $\varphi^{-1}: \varphi[K] \rightarrow K$,

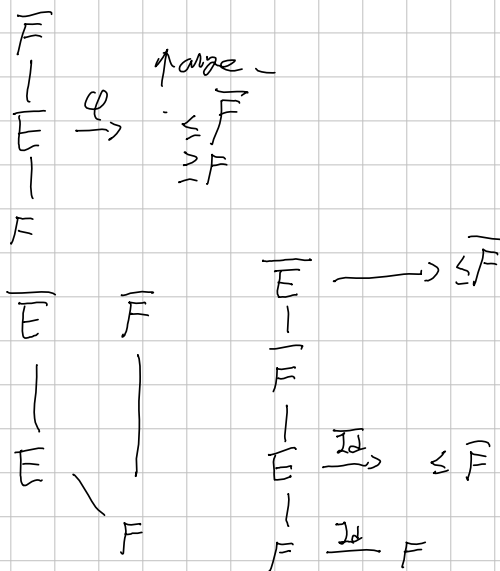
$$\overline{\varphi^{-1}}: K \rightarrow \overline{K} = K \text{ s.t. } \overline{\varphi^{-1}}|_{\varphi[K]} = \varphi^{-1}.$$

Then, $\overline{\varphi^{-1}}[K] \subseteq K$ and $\overline{\varphi^{-1}}[\varphi[K]] = \varphi^{-1}[\varphi[K]] = \varphi^{-1}[\varphi[K]] \supseteq K$.

Thus, $\overline{\varphi^{-1}}[K] = K$ Now,

$$K = (\overline{\varphi^{-1}})^{-1}[K]$$

$$\begin{array}{ccc} \overline{K} & \xleftarrow{\overline{\varphi^{-1}}} & K \\ \parallel & & \downarrow \\ K & \xrightleftharpoons[\varphi^{-1}]{\varphi} & K' \end{array}$$



$$\Gamma(\overline{\mathbb{Q}(\sqrt{2})}) \cong \overline{\mathbb{Q}(\sqrt{2})}.$$

Let $\varphi: \mathbb{Q}(\sqrt{2}) \rightarrow \mathbb{Q}(\sqrt{2})$ be an Isomorphism
 And, $\overline{\varphi}: \overline{\mathbb{Q}(\sqrt{2})} \rightarrow \overline{\mathbb{Q}(\sqrt{2})}$ be an Extension.

$$\text{ETS; } \overline{\varphi}[\overline{\mathbb{Q}(\sqrt{2})}] = \overline{\mathbb{Q}(\sqrt{2})}$$

$$\overline{F(x)} = \overline{F(y)} \quad ? \quad \overline{\mathbb{Q}(x)} = \overline{\mathbb{Q}(y)} \quad ?$$

Thm. If E/F is finite extension, then $\{E:F\} \leq [E:F]$

Proof. Let $E = F(\alpha_1, \dots, \alpha_n)$ where α_i is algebraic over F .

For each $1 \leq k \leq n-1$,

$$\{F(\alpha_1, \dots, \alpha_k, \alpha_{k+1}) : F(\alpha_1, \dots, \alpha_k)\} \leq [F(\alpha_1, \dots, \alpha_{k+1}) : F(\alpha_1, \dots, \alpha_k)]$$

Because the number of extensions is determined by degree of

$$\text{irr}(\alpha_{k+1}, F(\alpha_1, \dots, \alpha_k)).$$

Thm. If E/F is finite extension and splitting field,

then E is a splitting field for some α over $F[x]$.

Proof. Since E is finite over F , denote $E = F(\alpha_1, \dots, \alpha_n)$ where α_i is algebraic over F .

Then, E is a splitting field for a polynomial

$$\text{irr}(\alpha_1, F) \cdot \dots \cdot \text{irr}(\alpha_n, F).$$

Because every root of $\text{irr}(\alpha_i, F)$ are contained in E .

If $[E:F] = 2$, then E is splitting field over F .

Proof. Let $\alpha \in E \setminus F$. Then, $\text{irr}(\alpha, F) = x^2 - ax + b$ where $a, b \in F$.

Since α is a root of this,

$$x^2 - ax + b = (x - \alpha)(x - \beta) \in \bar{F}[x]$$

Consider $x^3 - 2$ in $\mathbb{Q}[x]$.

$x^3 - 2$ have three roots in $\mathbb{C}$ as

$$\sqrt[3]{2}, \sqrt[3]{2} \cdot \frac{-1 + i\sqrt{3}}{2}, \sqrt[3]{2} \cdot \frac{-1 - i\sqrt{3}}{2}$$

Determine the splitting field of $x^3 - 2$ is:

$$\mathbb{Q}(\sqrt[3]{2}, i\sqrt{3})$$

Because,

$$x^3 - 2 = (x - \sqrt[3]{2}) \underbrace{\left(x - \sqrt[3]{2} \cdot \frac{-1 + i\sqrt{3}}{2} \right) \left(x - \sqrt[3]{2} \cdot \frac{-1 - i\sqrt{3}}{2} \right)}_{\sqrt[3]{2} \cdot \varphi_3}$$

$$\begin{array}{ccccc} & & \mathbb{Q}(\sqrt[3]{2}, \varphi_3) = \mathbb{Q}(\sqrt[3]{2}, i\sqrt{3}) & & \\ & \swarrow 2 & | 3 & \searrow 2 & \\ \mathbb{Q}(\sqrt[3]{2}) & & \mathbb{Q}(i\sqrt{3}) & & \mathbb{Q}(\varphi_3) \\ & \searrow 3 & | 2 & \swarrow 3 & \\ & & \mathbb{Q} & & \end{array}$$

$\mathbb{Q}(i\sqrt{3})(\sqrt[3]{2})$ is a splitting field for $x^3 - 2 \in \mathbb{Q}(i\sqrt{3})[x]$

Therefore $|\text{Aut}(\mathbb{Q}(i\sqrt{3}, \sqrt[3]{2}) / \mathbb{Q}(i\sqrt{3}))| = 3$.

Thm. Let E/F be a splitting field of $f(x) \in F[x]$.

If $\sigma: E \rightarrow E$ is Automorphism leaving F fixed,

then for any root $\alpha \in E$ of $f(x)$, $\sigma(\alpha) \in E$ is root of $f(x)$.

proof. Denote $f(x) = a_0 + a_1x + \dots + a_nx^n$. Since $f(\alpha) = 0$,

$$0 = \sigma(f(\alpha)) = \sigma(a_0 + a_1\alpha + \dots + a_n\alpha^n) = a_0 + a_1\sigma(\alpha) + \dots + a_n\sigma(\alpha)^n = f(\sigma(\alpha))$$

Thm. Let E/F be a splitting field of $f(x) \in F[x]$.

Then, $\text{Aut}(E/F) \cong S_R$ where R is distinct root of $f(x)$ in E .

E/F .

$$|\text{Aut}(E/F)| \underset{\substack{\uparrow \\ E \text{ splitting } F}}{=} \{E:F\} \underset{\substack{\uparrow \text{ separable } \Rightarrow}}{\leq} [E:F]$$

Let K be an algebraically closed field.

$\varphi: K \rightarrow K' (\subseteq K)$ is $\exists$, where K is algebraic over K' ,
then φ is one-to-one .

proof. $\varphi^{-1}: K' \rightarrow K$

$$\varphi^{-1}: K \rightarrow K'$$

$\downarrow$ Extension

$$(\overline{\varphi^{-1}})^{-1}: K \rightarrow K.$$

$$\overline{\varphi^{-1}}: K \rightarrow \overline{K} = K.$$

$$b = \varphi(a) \in K'$$

If $K \setminus K' \neq \emptyset$, $a \in K \setminus K'$. Then, $\exists b \in K'$ s.t. $\varphi^{-1}(b) = a$.

$$\exists c \in K \text{ s.t. } \overline{\varphi^{-1}}(c) = a.$$

$$\varphi(\overline{\varphi^{-1}}(a)) \in K'$$

$$\text{Then, } \overline{\varphi^{-1}}(b) = \overline{\varphi^{-1}}(\varphi(a)) = a$$

$$\varphi: K \rightarrow K' \text{ Iso.}$$

$$\varphi^{-1}: K' \rightarrow K.$$

$\downarrow$ Ext.

$$\overline{\varphi^{-1}}: K \rightarrow \overline{K} = K.$$

$$\forall x \in K, \overline{\varphi^{-1}}(\varphi(x)) = \varphi^{-1}(\varphi(x)) = x.$$

$$\text{That is, } \overline{\varphi^{-1}}[\varphi[K]] = K.$$

Thm. Let F be a finite field of characteristic p .

Then, a map

$$\mathcal{Q}_p: F \rightarrow F: x \mapsto x^p$$

is Automorphism. Moreover, $F_{\{\mathcal{Q}_p\}} = \mathbb{Z}_p$.

Proof. Well-defined is clear.

Homomorphism: for any $a, b \in F$,

$$\mathcal{Q}_p(a+b) = (a+b)^p = \sum_{\substack{k=0 \\ \text{binomial}}}^p \binom{p}{k} a^{p-k} b^k = a^p + b^p = \mathcal{Q}_p(a) + \mathcal{Q}_p(b)$$

$\uparrow$ $\text{ch}(F)=p$

And clearly preserves multiply.

Injectivity:

$$\mathcal{Q}_p(a) = a^p = 0 \Leftrightarrow a = 0 \Rightarrow \ker \mathcal{Q}_p = \{0\}.$$

Surjectivity: Finite injective endomorphism, thus surjective.

Meanwhile, by the Fermat's Theorem, for any $a \in \mathbb{Z}_p$, (Isomorphically the prime field of F)

$$\mathcal{Q}_p(a) = a^p = a$$

Thus $\mathbb{Z}_p \subseteq F$. And, observe that

$$\mathcal{Q}_p(x) = x \Leftrightarrow x^p = x \Leftrightarrow x^p - x = 0.$$

This polynomial has at most p zeros in F , thus $\mathbb{Z}_p = F_{\{\mathcal{Q}_p\}}$.